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Editors-in-Chief
Smart Agricultural Technology

Dear Editors,

Please consider the manuscript, “Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues,” for publication in *Smart Agricultural Technology*.

The study addresses an applied quality-assurance question for agricultural vision. When only a finite number of predicted regions can be reviewed, how much distinct plant support survives candidate formation, role qualification, ranking, and budget truncation? We organise these stages into an executable evaluation contract for offline annotation quality assurance and model-failure review.

The contract separates spatial support, role-qualified support, merge-aware set coverage, and one-to-one support. The two final endpoints bound geometric recoverability when one region creates one quality-assurance record. On the official WeedsGalore split, proposal operating-point commissioning expands raw support, while a 27-contract analysis shows that queued effects also depend on the proposal distribution used to train the ranker.

The model study compares U-Net, DeepLabV3+, and Mask R-CNN under the same $K = 20$ capacity. It includes crossed training-seed and test-image uncertainty, standard AP and AR, a queue-level PQ-style diagnostic, contract-rank stability, and crop-exposure sensitivity. Mask R-CNN provides the highest internal point estimate, while the crossed interval for its contrast with U-Net includes zero.

Two independent operators reviewed a frozen 518-region queue. Their agreement is reported with Cohen’s κ , Gwet’s AC1, weighted κ , confidence distributions, and complete aggregate contingency tables. Human reviewability and geometric qualification are reported as distinct evidence sources.

A separately locked transfer applied five validation-selected U-Net checkpoints to 60 CWFID images without external fitting. No checkpoint formed a candidate queue. Post-lock continuous-score diagnostics preserve this primary result and show weak, checkpoint-variable ranking and unstable low-threshold components. The study therefore defines an auditable transfer boundary rather than a deployment claim.

The accompanying GitHub release contains figure source data, frozen configurations, per-image and per-seed outputs, matching tables, model and data hashes, preserved failures, and replay programs. The compact verifier reproduces archived tables and numerical invariants. Full retraining is supported through separate public-data and model-acquisition instructions.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors.

Sincerely,

Zhuo Chen
on behalf of all authors